Construction Chemicals

SUBMITTAL DOCUMENT

Mixcoat Novolac

Solvent-Free Novolac Epoxy Coating
for Chemical-Resistant Floors and Linings



Chemical Resistant	Solvent Free	High Strength	Abrasion
67 reagents rated	> 98 % solids	> 75 N/mm²	55 mg loss

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

Mixcoat Novolac — Solvent-Free Novolac Epoxy Coating for Chemical-Resistant Floors and Linings

Project Name		**Document No.**	OMX-MS-MIXCOATNOVOLAC
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Mixcoat Novolac	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Mixcoat Novolac on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers surface preparation, mixing, application and quality control of Mixcoat Novolac, a solvent-free novolac epoxy coating, for heavy-duty chemical-resistant linings of concrete and steel in dairies, food and beverage plants, hospitals, pharmaceutical and chemical plants, sugar refineries, battery rooms, wastewater plants and containment areas.

- Inspection and acceptance of the substrate and the work area.
- Surface preparation and repair of defects.
- Priming of the prepared surface.
- Mixing and application of the material as described in this document.
- Curing, protection and handover of the completed area.
- Quality control, inspection and documentation of each stage.

3. Definitions

Term	Meaning
Pot life	Time the mixed material remains workable after Part A and Part B are combined.
Recoat interval	Minimum and maximum time between coats to achieve inter-coat adhesion.
WFT / DFT	Wet film thickness measured during application / dry film thickness after cure.
Dew point	Temperature at which moisture condenses; the substrate must be at least 3 °C above it.
Kit / set	Pre-weighed Part A and Part B supplied to be mixed completely.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
Mixcoat Novolac	Novolac epoxy coating, 5 / 20 kg A + B sets
Epoxy primer	O-Prime SF or O-Prime ST
Epoxy filler	O-Putty for cracks and honeycombs
Quartz sand	Graded, dry, for anti-slip broadcast

5.1 Key product data

Property	Typical Value
Mixing ratio (A : B)	2 : 1 by volume — pre-weighed sets, mix full units as recommended
Colour	Natural, white, grey; other colours on request
Mixed density	1.5 ± 0.1 g/cm³
Volume solids	> 98 %
Pot life	30 – 40 min
Touch dry	6 – 8 hours
Recoat interval	Min. 16 hours / max. 48 hours
Initial cure	24 hours

6. Tools & Equipment

- Grinder / shot blaster
- Vacuum
- Low-speed mixer (300–400 rpm) with helical paddle
- Short-nap rollers

- Brushes
- Airless spray
- Wet-film comb
- Holiday detector
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Wet-film thickness gauge
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Organic-vapour respirator
- Nitrile gloves
- Goggles
- Overalls

7. Storage & Handling

- 12 months in original, unopened packaging. Store dry and covered at +10 °C to +30 °C, away from heat, sunlight and frost.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep Part A and Part B of each kit together and do not mix parts from different batches.
- Store away from heat, sparks and ignition sources.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Concrete must be at least 28 days old, sound, dry and fully cured.
2. Scarify, shot blast or grind to remove laitance, loose aggregate, efflorescence, release agents and old coatings.

3. Open and fill large cracks and honeycombs with epoxy filler; seal hairline cracks with the primer coat.
4. Measure substrate temperature, air temperature and relative humidity; confirm the substrate is at least 3 °C above dew point.
5. Protect adjacent surfaces, drains and fixtures with masking tape and sheeting.
6. Obtain QA/QC approval of the prepared surface before starting the next step.

10. Mixing

1. Condition materials to between +15 °C and +25 °C before mixing.
2. Check the batch numbers and expiry dates of both parts and record them.
3. Empty the full Part B into the Part A container (2 : 1 by volume).
4. Mix 3 minutes at 300–400 rpm with a helical paddle, scraping sides and bottom.
5. Do not split kits unless weighed on calibrated scales; do not thin.
6. Mix only the quantity that can be applied within the pot life.

11. Application

1. Apply the primer at 0.15 – 0.20 kg/m² and allow to cure.
2. Apply Mixcoat Novolac evenly by short-nap roller, brush or airless spray, working continuously.
3. For anti-slip finish, broadcast quartz sand into the first coat within 15 minutes; remove excess after cure.
4. Apply the second coat within the recoat window (16 – 48 hours); if exceeded, abrade first.
5. Work in manageable bays, keeping a wet edge to avoid lap marks.
6. Check consumption by area and wet film thickness regularly during application and record the results.

12. Curing & Protection

- Pot life 30 – 40 minutes at 25 °C; discard material that starts to gel.
- Touch dry 6 – 8 hours; initial cure 24 hours.
- Full chemical resistance after 7 days at 25 °C.
- Protect the applied area from water, dust, other trades and traffic until the stated cure times are reached.
- Barricade the area and display warning signs.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Substrate	Before priming	≥ 28 days, sound, moisture < 4 %	QA/QC
Climate	During work	+10 °C to +35 °C, RH ≤ 85 %, 3 °C above dew point	Site Engineer
Film thickness	During application	WFT checked with comb to meet specified DFT	QA/QC
Continuity	After cure	Holiday test passed; no pinholes, bubbles or crawling	QA/QC

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point	Start of shift and every 4 hours	Daily climate log
Surface preparation	Visual, moisture, soundness	Each area	Inspection request (hold point)
Application	Consumption, thickness, appearance	Each coat / area	Daily progress report
Final inspection	Visual, continuity, finish	Each completed area	Handover checklist

14. Defects & Remedies

Defect	Likely Cause	Remedy
Pinholes / bubbles	Outgassing from porous substrate; air entrapped in mixing	Seal with primer; mix at low speed; de-aerate with spiked roller; sand and recoat
Blushing / tacky surface	High humidity or low temperature during cure	Wash and abrade; recoat in suitable conditions
Poor adhesion	Contaminated, weak or damp substrate; recoat window exceeded	Remove, re-prepare surface and reapply
Soft spots	Incorrect mixing or ratio	Remove affected area and reapply with full kits
Uneven colour / lap marks	Loss of wet edge; different batches	Abrade and apply a further coat from one batch

15. Health, Safety & Environment

Hazard	Control Measures
Skin and eye contact	Gloves, goggles, overalls; eye-wash available
Inhalation of vapours / dust	Ventilation; masks as required
Slips and trips	Housekeeping; barricades and signs
Manual handling	Correct lifting technique; mechanical aids
Electrical tools	Inspected tools; protected cables

- Wear nitrile gloves, goggles and overalls; use an organic-vapour mask when spraying.
- Uncured epoxy may cause skin sensitisation.
- Ensure good ventilation in confined spaces.
- Eyes: rinse with water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean tools with epoxy thinner before the material cures; cured material can only be removed mechanically.

17. Limitations

- Ambient and substrate: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Do not apply in rain or when rain is expected before the material has set.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Concrete age ≥ 28 days verified	☐ Yes ☐ No	
Surface profiled and vacuumed	☐ Yes ☐ No	
Cracks and honeycombs repaired	☐ Yes ☐ No	
Primer applied and cured	☐ Yes ☐ No	
Full kits mixed 3 min at 300–400 rpm	☐ Yes ☐ No	
Climate conditions within limits	☐ Yes ☐ No	
WFT checked per coat	☐ Yes ☐ No	
Second coat within 16 – 48 hours	☐ Yes ☐ No	
Holiday test and visual inspection passed	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

Mixcoat Novolac — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Protective Coating

Mixcoat Novolac

Solvent-Free Novolac Epoxy Coating
for Chemical-Resistant Floors and Linings



Product Description

Mixcoat Novolac is a two-component, solvent-free, low-viscosity novolac epoxy coating for aggressive chemical environments. It gives a durable, chemical-resistant and slip-resistant finish on prepared concrete and steel, and can also be used as a binder in resin mortar and slurry systems.

Uses

- Heavy-duty coating for dairies.
- Food processing and beverage plants.
- Hospitals and pharmaceutical industry.
- Chemical industry and sugar refineries.
- Power stations and battery rooms.
- Heavy engineering workshops.
- Internal protection of concrete tanks containing chemicals.
- Sewage and wastewater treatment plants and liquid containment areas.

Advantages

- Solvent-free — no fire risk, no shrinkage on cure.
- Excellent adhesion to concrete and steel.
- Anti-slip, anti-skid finish option.
- High mechanical strength.
- High resistance to abrasion.
- High chemical resistance.

Application

1. **Surface Preparation**
   Concrete ≥ 28 days old, sound and dry (< 4 % moisture). Remove laitance and old coatings mechanically.
2. **Priming**
   Apply epoxy primer at 0.15 – 0.20 kg/m² to seal hairline cracks and pores. Allow to dry.
3. **Top Coat**
   Mix full A + B kit (2:1 by volume) for 3 min. Apply 2 coats by roller, brush or airless spray.

Consumption

0.50 – 0.60 kg/m² (2 coats, smooth)
Textured 0.8 – 1.0 · Anti-slip 1.6 – 2.0 kg/m²
Full system table on page 2.

Packaging

5 kg and 20 kg sets: Part A in pail + Part B in jerrycan.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry and covered at +10 °C to +30 °C, away from heat, sunlight and frost.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	2 : 1 by volume — pre-weighed sets, mix full units as recommended	—
Colour	Natural, white, grey; other colours on request	—
Mixed density	1.5 ± 0.1 g/cm³	ASTM D1475
Volume solids	> 98 %	—
Pot life	30 – 40 min	—
Touch dry	6 – 8 hours	—
Recoat interval	Min. 16 hours / max. 48 hours	—
Initial cure	24 hours	—
Full cure	7 days	—
Compressive strength	> 75 N/mm²	ASTM D695
Flexural strength	> 50 N/mm²	ASTM D790
Tensile strength	> 30 N/mm²	ASTM D638
Pull-off adhesion to concrete	> 2.5 N/mm² (concrete failure)	ASTM D4541
Abrasion resistance (CS-17, 1000 g, 1000 cycles)	55 mg loss	ASTM D4060
Hardness	≥ 80 Shore D	ASTM D2240
Dry heat resistance	Up to +120 °C	—

Consumption

System	Coats	Consumption	DFT
Primer (epoxy primer)	1	0.15 – 0.20 kg/m²	—
Smooth finish	2	0.50 – 0.60 kg/m²	0.3 – 0.4 mm
Textured finish	2	0.80 – 1.00 kg/m²	0.5 – 0.6 mm
Anti-slip finish (quartz broadcast)	2	1.60 – 2.00 kg/m²	1.0 – 3.0 mm (incl. quartz)

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Substrate moisture: < 4 %.
- Full chemical resistance: 7 days at 25 °C (10 days at 15 °C).

Application Notes

- Stir Part A, add all of Part B, mix at 300–400 rpm.
- Pour in a line and spread with a nap-shed resistant roller; work continuously.
- If the recoat window is missed: solvent wipe, abrade, solvent wipe.
- Broadcast aggregate within 15 minutes if anti-slip is required.
- Do not thin. Discard material that starts to gel.

Chemical Resistance

Reagent	Rating	Reagent	Rating
Acetic Acid 20%	3	Isophorone	5
Acetone	3	Iso-propanol	5
Ammonia 40%	5	Jet Fuel	3
Benzene	3	Kerosene	5
N-Butanol	5	Lactic Acid	5
N-Butyl Acetate	5	Linseed Oil	5
Calcium Hydroxide 30%	5	Methanol	5
Castor Oil	5	Methylene Chloride	0
Chlorinated Paraffin	5	Methyl Ethyl Ketone (MEK)	3
Chloroform	3	Methyl Isobutyl Ketone (MIBK)	3
Citric Acid 30%	5	Milk	5
Crude Oil	5	Mineral Oils	5
Cyclohexanone	5	Naphtha	5
Deionised / Distilled Water	5	Nitric Acid 10%	5
Diacetone Alcohol	5	Nitric Acid 30%	3
Diesel Oil	5	Nitric Acid 69%	0
Di-isobutyl Ketone	5	Nonyl Phenol	5
Dioxane	3	Phenol	5
Dipentene	5	Phosphoric Acid 10%	5
Epichlorohydrin	1	Phosphoric Acid 20%	5
Ethyl Acetate	5	Phosphoric Acid 50%	3
Ethyl Benzene	3	Potassium Hydroxide	5
Ethylene Glycol	5	Sea Water	5
Ethylene Glycol Monoethyl Ether	5	Sodium Hydroxide	5
Formaldehyde	5	Styrene	3
Formic Acid	5	Sugar Solution 30%	5
Furfural	3	Sulphuric Acid 10%	5
Glycerol	5	Sulphuric Acid 50%	3
Hydrazine	3	Sulphuric Acid 98%	0
Hydrochloric Acid 35%	3	Toluene	5
Hydrogen Peroxide	5	Toluene Diisocyanate	5
Hydrogen Sulphide	3	White Spirit	5
Iso-butanol	5	Xylene	5
Iso-butyl Acetate	5		

Rating after 7 days cure at 25 °C: 5 = excellent (long-term) 3 = medium term (1 month) 1 = short term (24 h) 0 = not resistant

Health & Safety

- Wear nitrile gloves, goggles and overalls; use an organic-vapour mask when spraying.
- Uncured epoxy may cause skin sensitisation.
- Ensure good ventilation in confined spaces.

First Aid & Cleaning

- Eyes: rinse with water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.
- Clean tools with epoxy thinner before the material cures; cured material can only be removed mechanically.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.